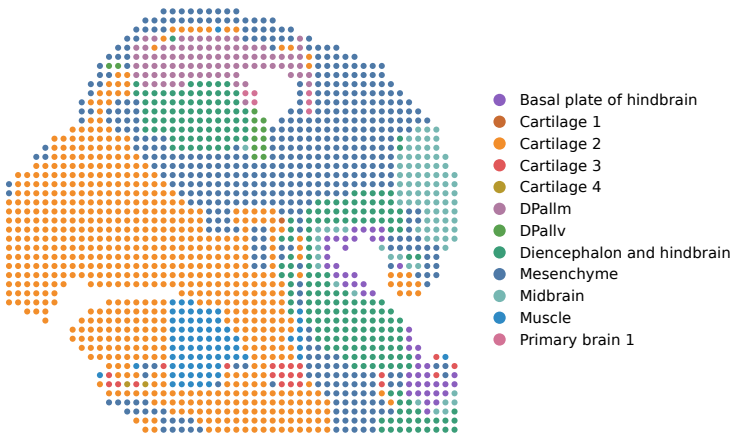
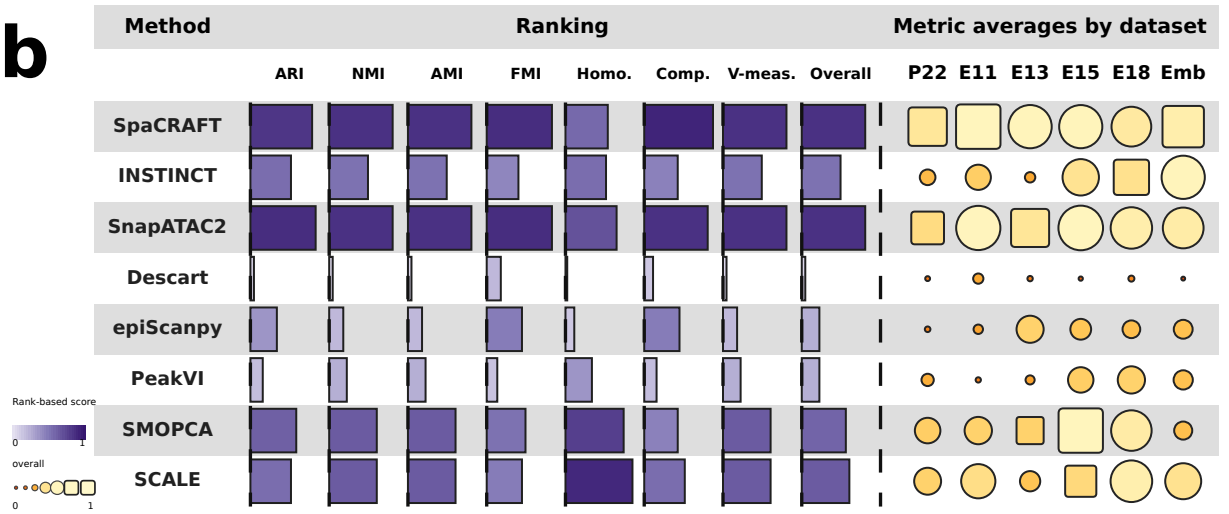
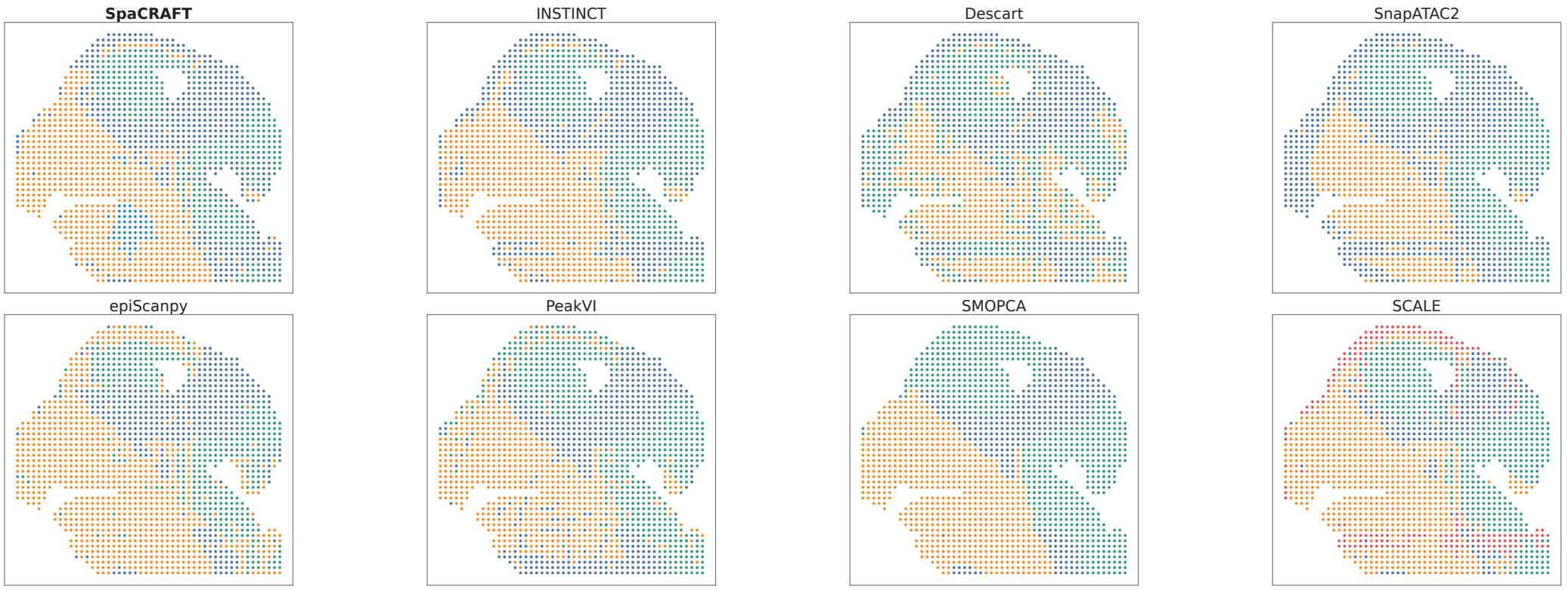
a


E13.5 ground truth
Basal plate of hindbrain
Cartilage 1
Cartilage 2
Cartilage 3
Cartilage 4
DPallm
DPallv
Diencephalon and hindbrain
Mesenchyme
Midbrain
Muscle
Primary brain 1

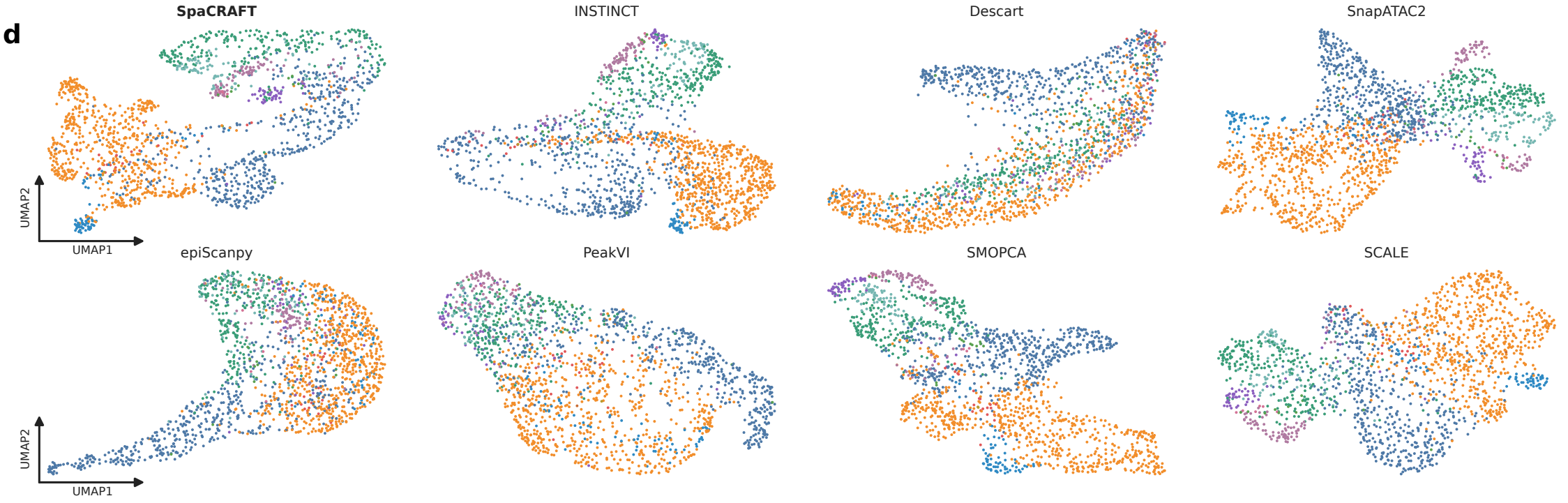
b


Method
Ranking
Metric averages by dataset
ARI
NMI
AMI
FMI
Homo.
Comp.
V-meas.
Overall
P22
E11
E13
E15
E18
Emb
SpaCRAFT
INSTINCT
SnapATAC2
Descart
epiScanpy
PeakVI
SMOPCA
SCALE
Rank-based score
0
1
overall
0
1

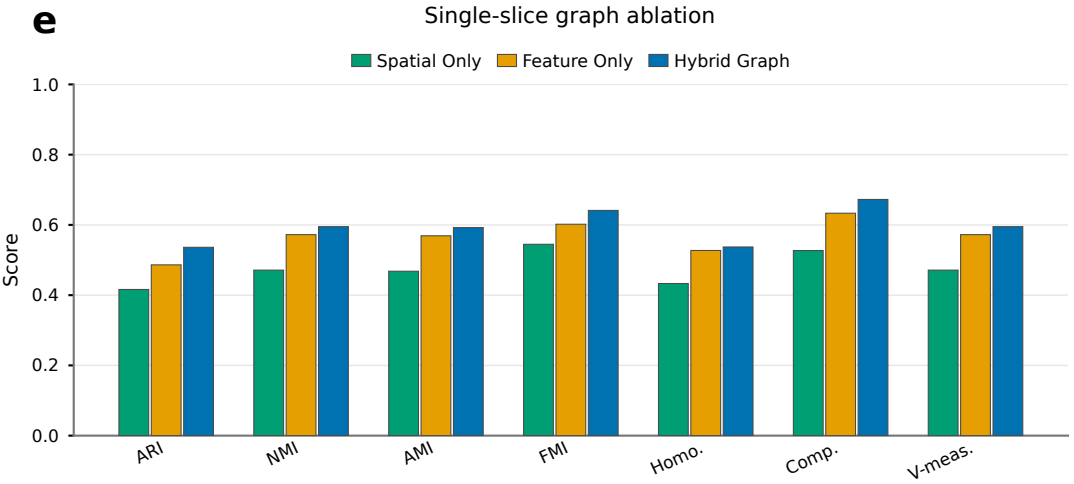
c

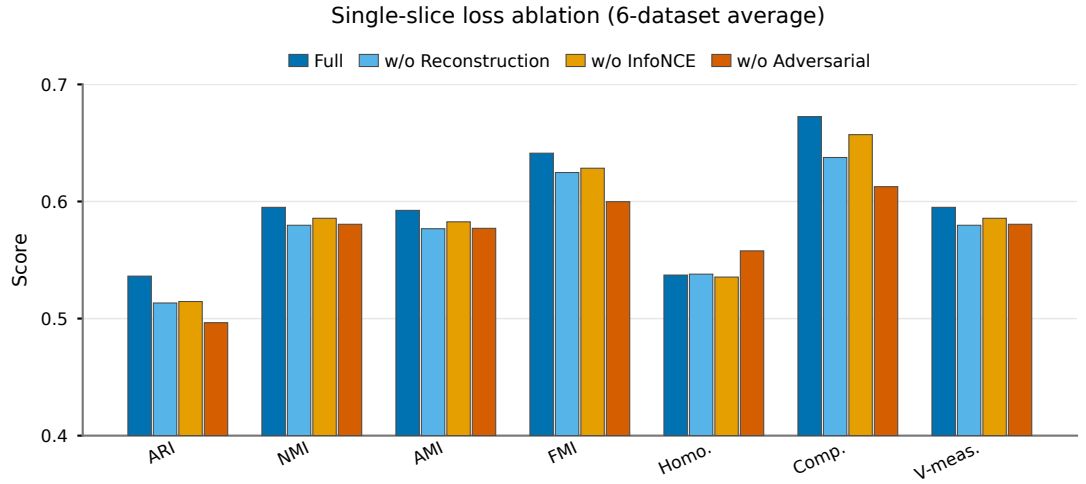

SpaCRAFT
INSTINCT
Descart
SnapATAC2
epiScanpy
PeakVI
SMOPCA
SCALE

d


SpaCRAFT
INSTINCT
Descart
SnapATAC2
epiScanpy
PeakVI
SMOPCA
SCALE
UMAP2
UMAP1
UMAP2
UMAP1

e


Single-slice graph ablation
Spatial Only
Feature Only
Hybrid Graph
Score
1.0
0.8
0.6
0.4
0.2
0.0
ARI
NMI
AMI
FMI
Homo.
Comp.
V-meas.


Single-slice loss ablation (6-dataset average)
Full
w/o Reconstruction
w/o InfoNCE
w/o Adversarial
Score
0.7
0.6
0.5
0.4
ARI
NMI
AMI
FMI
Homo.
Comp.
V-meas.